APPENDIX 1: Search syntax

PUBMED SEARCH
"Otitis Media"[Mesh] OR otitis media[Title/Abstract] OR ear infection*[Title/Abstract] OR ear inflammation*[Title/Abstract] OR otiti* media[Title/Abstract] OR middle ear inf*[Title/Abstract] OR OME[Title/Abstract] OR OMA[Title/Abstract] OR CSOM[Title/Abstract] OR otitides[Title/Abstract] OR ACSOM[Title/Abstract] AND "Hearing Loss"[Mesh] OR hearing loss*[Title/Abstract] OR deafness[Title/Abstract] OR cochlear dysfunction*[Title/Abstract] OR hypoacusis[Title/Abstract] OR hearing impairment[Title/Abstract] OR inner ear impairment[Title/Abstract]
EMBASE SEARCH
'otitis media'/exp OR 'otitis media':ti,ab,kw OR 'ear infection*':ti,ab,kw OR 'ear inflammation*':ti,ab,kw OR 'otiti* media':ti,ab,kw OR 'middle ear inf*':ti,ab,kw OR 'OME':ti,ab,kw OR 'OMA':ti,ab,kw OR 'CSOM':ti,ab,kw OR 'otitides':ti,ab,kw OR 'ACSOM':ti,ab,kw AND 'inner ear hearing loss'/exp OR 'hearing loss*':ti,ab,kw OR 'deafness':ti,ab,kw OR 'cochlear dysfunction*':ti,ab,kw OR 'hypoacusis':ti,ab,kw OR 'hearing impairment':ti,ab,kw OR 'inner ear impairment':ti,ab,kw
COCHRANE SEARCH
"otitis media":ti,ab,kw OR "ear infection*":ti,ab,kw OR "ear inflammation*":ti,ab,kw OR "otiti* media":ti,ab,kw OR "middle ear inf*":ti,ab,kw OR "OME":ti,ab,kw OR "OMA":ti,ab,kw OR "CSOM":ti,ab,kw OR "otitides":ti,ab,kw OR "ACSOM":ti,ab,kw AND "hearing loss*":ti,ab,kw OR "deafness":ti,ab,kw OR "cochlear dysfunction*":ti,ab,kw OR "hypoacusis":ti,ab,kw OR "hearing impairment":ti,ab,kw OR "inner ear impairment":ti,ab,kw

APPENDIX 2: predefined risk of bias consistently criteria

The Quality In Prognosis Studies tool (QUIPS)

Domain	Criteria
Study Participation	- In-/exclusion criteria: if only including patients with audiograms, choose no (selection bias). - Baseline characteristics: when most important factors are described but no table is given, choose yes.
Study attrition	No additional criteria
Prognostic factor measurement	- Definition of the prognostic factor: must contain >3 months.
Outcome measurement	No additional criteria
Study confounding	- Confounders: age, duration of disease, (prior) use of ototoxic ear drugs.
Statistical analysis and reporting	No additional criteria

Newcastle-Ottawa Score (NOS)

Category	Criteria
• Selection	No additional criteria
• Comparability	- Total of 1 star when accounting for age (cohort: contralateral ear; case-control: matching). - Total of 2 stars when also accounting for duration of disease and prior use of ear drugs.
• Outcome	- Choose record linkage when outcome information was derived from chart reviews. - Follow up was long enough when duration of disease was >3 months.